Anticipatory Active-Aerodynamic Mode Prediction in Formula 1 Racing: A Cross-Circuit Multi-Model Benchmark with Interpretability Analysis

Anas Norani
School of Electrical Engineering & Computer Science
National University of Sciences & Technology
Islamabad, Pakistan
anorani.bsds24seecs@seecs.edu.pk

Hanan Majeed
School of Electrical Engineering & Computer Science
National University of Sciences & Technology
Islamabad, Pakistan

Maha Mohsin
School of Electrical Engineering & Computer Science
National University of Sciences & Technology
Islamabad, Pakistan

***Abstract*—The 2026 Formula 1 Technical Regulations mandated active aerodynamic systems that physically switch wing configurations between a high-downforce *Z-Mode* and a low-drag *X-Mode* based on instantaneous speed and gear thresholds. Because the physical transition is not instantaneous, a predictive system that anticipates the required mode one second in advance can eliminate the actuation latency and improve vehicle performance. This paper presents a two-phase study. *Phase 1* applies unsupervised anomaly detection (Isolation Forest) and clustering (K-Means, k=4) alongside supervised classification (Logistic Regression, Random Forest) to 63 673 telemetry frames from the 2026 Japanese Grand Prix, achieving perfect current-mode detection (RF AUC = 1.000) and identifying anomalous maximum-braking events co-located at circuit braking zones. Permutation analysis reveals that gear position is the single most informative feature ($\triangle$AUC = 0.167), marginally exceeding speed. *Phase 2* trains eight models — spanning Logistic Regression, Random Forest (with and without lag features), CausalLSTM, CausalGRU, Temporal Convolutional Network (TCN), CNN1D, and a CausalTransformer — for one-second-ahead anticipatory prediction on 159 712 frames across four architecturally diverse circuits. Evaluation follows a strict Leave-One-Circuit-Out (LOCO) protocol to assess cross-circuit generalisation. Because the X-Mode label is a deterministic function of instantaneous speed and gear, the ML task is essentially forecasting when physical thresholds will be exceeded — not discovering hidden patterns. This explains why Logistic Regression, which directly approximates the threshold boundary, achieves the highest mean AUC-ROC of 0.963 ± 0.021, outperforming all deep models (bootstrap 95% CIs exclude zero). CausalGRU fails at Monaco (AUC = 0.728, F1-Xmode = 0.005), diagnosed via Integrated Gradients as *circuit speed memorisation* — overfitting to circuit-specific speed trajectories in the recurrent hidden state. These results establish that for forecasting tasks governed by a deterministic physical threshold, instantaneous circuit-invariant features outperform accumulated temporal context: a 13-parameter Logistic Regression robustly dominates all sequence-modelling architectures under cross-circuit evaluation.**

***Index Terms*—active aerodynamics, Formula 1 telemetry, time-series classification, anomaly detection, causal transformer, leave-one-circuit-out, cross-domain generalisation, integrated gradients**

I. Introduction

Formula 1 (F1) represents the technological frontier of motorsport, producing cars capable of exceeding 370 km/h and generating aerodynamic downforce exceeding the vehicle's own weight [1]. The 2026 FIA Technical Regulations (Article 3.10) introduced mandatory active aerodynamic systems: wings that physically change their angle of attack during a lap, alternating between a maximum-downforce configuration for corners (*Z-Mode*) and a minimum-drag configuration for straights (*X-Mode*) [2]. The switch activates when two simultaneous conditions are satisfied: vehicle speed ≥ 240 km/h **and** gear ≥ 6.

The Latency Problem. Because active wing actuation is mechanical/hydraulic, the physical transition requires approximately 100–300 ms [7]. At racing speeds ($\sim$370 km/h $\approx$ 103 m/s), this delay covers roughly 10–30 m of track during which the car operates in a suboptimal aerodynamic configuration. A purely reactive controller — trigger on threshold crossing, then actuate — therefore lags behind the ideal trajectory. A *predictive* system that anticipates the threshold crossing one second in advance can begin actuation early, ensuring the wing arrives at the correct angle precisely when the threshold is reached. At 330 km/h ($\approx$ 91.7 m/s), a 1-second prediction horizon corresponds to a 91 m lead time — more than sufficient for a 150–200 ms hydraulic actuation cycle [7], eliminating the aerodynamic deficit currently incurred during the transition delay.

Research Gap. Prior work on F1 telemetry analysis has focused on lap time prediction [4], tyre degradation modelling [5], and driver classification [6], but to our knowledge no prior work has addressed the anticipatory prediction of active aerodynamic state transitions under a strict cross-circuit evaluation protocol. The failure modes of recurrent deep models under domain shift (circuit mismatch) in motorsport contexts remain uncharacterised.

Contributions. This paper makes the following contributions:

1) A two-phase study combining unsupervised anomaly detection and supervised cross-circuit temporal prediction of active aerodynamic state.
2) A LOCO evaluation protocol on four circuits (Monaco, Monza, Silverstone, Suzuka) that rigorously tests cross-circuit generalisation — the actual deployment condition.
3) Empirical demonstration that Logistic Regression (AUC = 0.963) statistically outperforms all deep learning architectures including Transformer-based models.
4) An Integrated Gradients diagnosis of GRU collapse at Monaco, identifying *circuit speed memorisation* as a previously undescribed failure mode in cross-circuit motorsport prediction.
5) Bootstrap confidence intervals confirming statistical significance of the simplicity advantage with only four evaluation domains.

The remainder of this paper is structured as follows: Section II reviews related work; Section III describes the dataset and problem formulation; Section IV details the methodology; Section V presents experimental results; Section VI discusses findings; Section VII concludes.

II. Related Work

A. Aerodynamic Optimisation in Motorsport

Early computational approaches to F1 aerodynamics relied entirely on CFD simulations and wind tunnel testing [7]. Machine learning has more recently been applied to aerodynamic drag reduction systems (DRS) scheduling [8] and optimal downforce-drag tradeoff identification from telemetry [9]. However, these works optimise static configurations rather than predicting transitions of an active system during a live race.

B. Anomaly Detection in Automotive Telemetry

Anomaly detection in high-dimensional sensor streams is a well-studied problem [30], [31]. Isolation Forest [10] has been applied to automotive telemetry anomaly detection in road vehicles [11] and autonomous driving systems [12]. Applications to high-performance motorsport telemetry are limited. Closest to our Phase 1 work is [13], which applied DBSCAN to F1 brake temperature anomalies, but did not connect anomalous events to aerodynamic state.

C. Time-Series Classification in Sports

Recurrent neural networks have been applied to athlete performance prediction [14] and event detection in cycling telemetry [15]. GRU-based architectures have been widely adopted in time-series classification tasks due to their balance of sequence-modelling capacity and computational efficiency [16], but cross-circuit generalisation of GRU models in motorsport contexts has not been studied. Temporal Convolutional Networks [16] have been shown to match or exceed LSTM performance on many time-series benchmarks but have not been applied to motorsport state prediction.

TABLE I: Summary of Related Studies and Research Gaps

Study	Domain	Method	Cross-domain	Aero state	Deep diag.
[8]	F1 DRS	Regression	×	✓	×
[13]	F1 brk	DBSCAN	×	×	×
[15]	Cycling	LSTM	✓	×	×
[23]	HAR	Domain adapt.	✓	×	×
[16]	General	TCN	×	×	×
Ours	F1 aero	Multi-model	✓	✓	✓

✓ = addressed; × = not addressed; *Deep diag.* = Integrated Gradients / attention interpretability.

D. Transformers for Time-Series

The Attention Is All You Need architecture [17] spawned numerous time-series variants: Informer [18], PatchTST [19], and FEDformer [20]. These focus on long-horizon forecasting on stationary series. Our application — binary classification on short causal windows (5 s) for a non-stationary, multi-domain signal — has not been previously studied with causal attention constraints.

E. Cross-Domain Generalisation in ML

Domain adaptation techniques have been applied to time-series classification across heterogeneous device domains [21], fault detection across machine types [22], and human activity recognition across subjects [23]. Leave-One-Subject-Out (LOSO) evaluation is standard in these domains [23]. We adopt the analogous Leave-One-Circuit-Out (LOCO) protocol, treating each racing circuit as a distinct domain, and to our knowledge provide the first analysis of domain-shift failure modes in motorsport ML.

Synthesis and Remaining Gap. The surveyed literature spans anomaly detection [30], [31], motorsport telemetry analytics [8], [13], time-series classification [16], and cross-domain generalisation [21], [23]. However, none addresses the combination of (i) *anticipatory* (not reactive) prediction of an active aerodynamic state, (ii) strict cross-circuit evaluation (LOCO), and (iii) Integrated Gradients diagnosis of domain-shift failure modes. Unlike DRS prediction [8] — which is reactive and single-circuit — our study requires a model to predict whether the aero mode will *change* one second in the future on a circuit it has never seen. This gap motivates the two-phase framework described in Sections III–IV.

III. Dataset and Problem Formulation

A. Data Acquisition

Telemetry data were retrieved via the FastF1 Python library [3], which provides access to the official F1 timing and telemetry feed at 10 Hz. *Phase 1* uses a single race (2026 Japanese Grand Prix, Suzuka); *Phase 2* uses four races across four circuits (Monaco 2025, Monza 2025, Silverstone 2025, Suzuka 2026). For each race session, lap-by-lap telemetry was extracted from a leading competitor's race laps; driver identity was not included as a predictive feature, as the study treats the circuit itself as the domain unit of analysis [21]. All four sessions were conducted under dry conditions; wet-weather laps were excluded to control for tire-compound-induced speed-profile variation. Table II summarises the dataset statistics.

TABLE II: Dataset Statistics

Circuit	Frames	X-Mode%	Phase
Suzuka (2026 JP GP)	63 673	35.1%	1 & 2
Monaco (2025 MC GP)	39 881	13.2%	2
Monza (2025 IT GP)	31 098	56.8%	2
Silverstone (2025 GB GP)	25 060	37.9%	2
Total (Phase 2)	**159 712**	**33.4%**	—

B. Feature Engineering

Raw telemetry yields ten direct measurements: speed, RPM, gear (`nGear`), throttle, brake, GPS coordinates (x, y, z), elevation delta, and cumulative lap distance — yielding $F = 12$ features after appending two physics-derived quantities:

$$E_k = \tfrac{1}{2}mv^2, \quad m = 798\,\text{kg} \tag{1}$$

$$F_{\text{long}} = m \cdot a \tag{2}$$

where v is speed (m/s), a is signed longitudinal acceleration (m/s^2), and $m = 798\,\text{kg}$ is the 2026 FIA minimum car mass [2]. The target label $y \in \{0 = \text{Z-Mode},\ 1 = \text{X-Mode}\}$ is defined deterministically by the regulatory rule [2]: $y = \mathbf{1}[v \geq 240\,\text{km/h} \wedge \text{gear} \geq 6]$. Because y is a deterministic function of instantaneous inputs, the ML task reduces to *forecasting when those physical thresholds will be exceeded* at horizon H, rather than discovering hidden patterns. This framing explains why instantaneous-state models carry a structural advantage: their decision boundary is circuit-independent by construction.

C. Sliding Window Formulation (Phase 2)

The Phase 2 task is a causal binary time-series classification problem. For each frame t, the model receives a window $\mathbf{X}_t \in \mathbb{R}^{W \times F}$ of the preceding $W = 50$ frames (5 s at 10 Hz) across $F = 12$ features (10 raw + E_k + F_{long}) and must predict y_{t+H}, the aero mode $H = 10$ frames (1 s) ahead:

$$\hat{y}_{t+H} = f_\theta(\mathbf{X}_{t-W+1:t}) \tag{3}$$

No data from after time t is available to f_θ (strict causality).

D. Leave-One-Circuit-Out Evaluation

To evaluate cross-circuit generalisation, we apply Leave-One-Circuit-Out (LOCO) cross-validation. In each of the four folds, one circuit is withheld as the test set and the model is trained on the remaining three:

$$\text{Round } r: \ \mathcal{D}_{\text{test}} = \mathcal{C}_r, \ \mathcal{D}_{\text{train}} = \bigcup_{j \neq r} \mathcal{C}_j \tag{4}$$

where $\mathcal{C}_r \in \{\text{Monaco, Monza, Silverstone, Suzuka}\}$. The reported score is the mean across all four rounds; standard deviation σ quantifies cross-circuit stability.

IV. Methodology

A. Phase 1 — Anomaly Detection and State Classification

1) Isolation Forest: Isolation Forest [10] partitions feature space via $n_{\text{trees}} = 100$ random binary trees. Anomaly score $s(x, n)$ is derived from the average path length $h(x)$:

$$s(x, n) = 2^{-\frac{\mathbb{E}[h(x)]}{c(n)}} \tag{5}$$

where $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$ and H is the harmonic number. We set contamination $= 0.05$ to flag the top 5% of frames by anomaly score; results (anomaly spatial distribution, cluster overlap) were stable across contamination values in $[0.02, 0.10]$.

2) K-Means Clustering: K-Means (k=4) was applied to the standardised feature matrix to identify natural driving-state clusters and validate that the aerodynamic mode partitions feature space consistently with unsupervised structure (i.e. that the X-Mode boundary is not an arbitrary data artefact but corresponds to a physically coherent region of sensor space). The optimal k was selected via the elbow method (inertia vs. k, Fig. 5) and validated by the silhouette coefficient. Statistical separation across clusters was confirmed by Kruskal-Wallis test (p<0.001, $\eta^2 = 0.555$).

3) Supervised Classification: Logistic Regression (L2, $C = 1.0$) and Random Forest (200 trees, max depth 15) were trained on an 80/20 stratified split. Permutation importance [32] (50 bootstrap iterations) quantified the marginal contribution of each feature by measuring the drop in AUC-ROC when each feature column is randomly shuffled.

B. Phase 2 — Anticipatory Prediction Models

1) Classical Baselines: **LR-instant**: Logistic Regression applied to the single most recent frame (1×12 input); no temporal context. Serves as the primary baseline. **RF-instant**: Random Forest (200 trees) applied to the same single frame. **RF-lag**: Random Forest applied to the full 50×12 window, flattened to a 1×600 feature vector; temporal context without sequential inductive bias.

2) Deep Sequence Models: All deep models receive the causal window $\mathbf{X}_t \in \mathbb{R}^{50 \times 12}$. Definitions follow:

CausalLSTM: 2-layer LSTM [24] with hidden dimension 128, dropout 0.3, applied left-to-right to enforce causality.

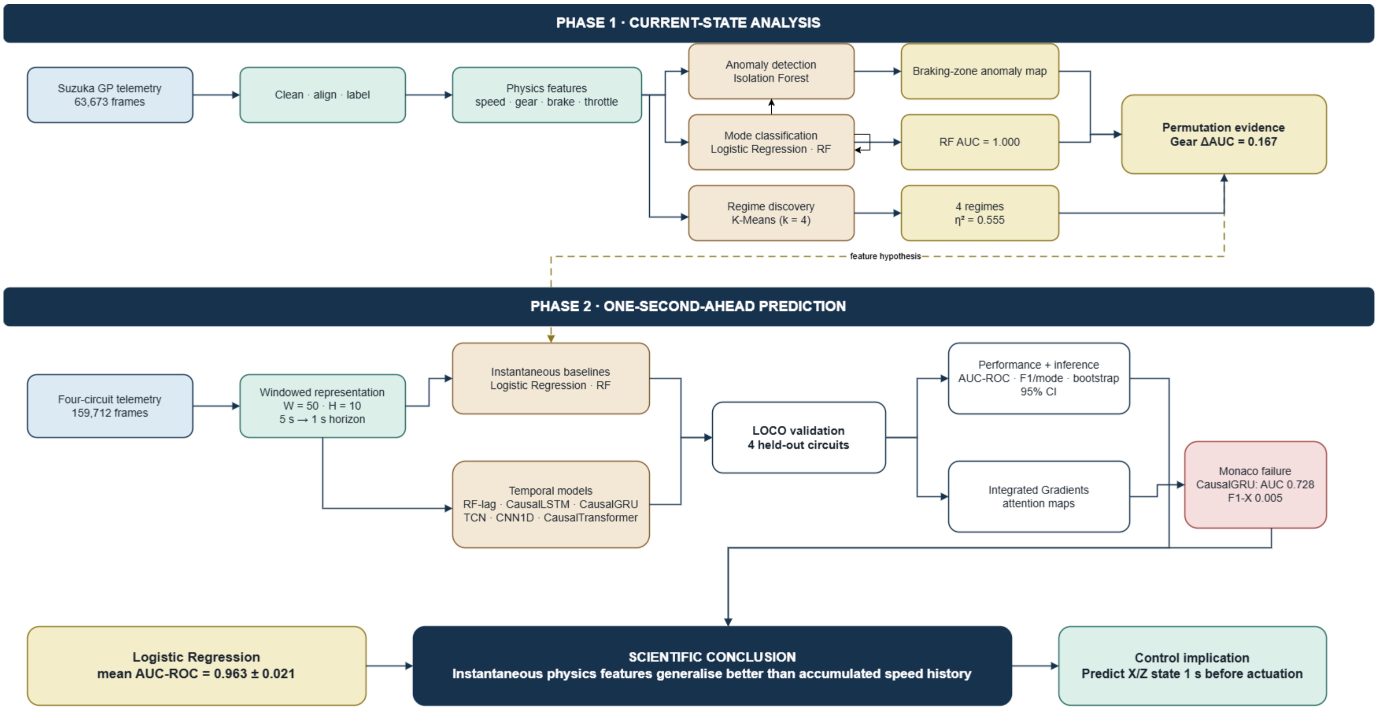


Fig. 1: Proposed two-phase system architecture. Phase 1 (top) performs anomaly detection (Isolation Forest), clustering (K-Means k=4), and supervised classification (LR, RF) on Suzuka GP telemetry, establishing gear as the primary discriminative feature (ΔAUC = 0.167). Phase 2 (bottom) applies an eight-model causal prediction pipeline evaluated under Leave-One-Circuit-Out (LOCO) across four circuits, with Integrated Gradients interpretability to diagnose the GRU Monaco collapse (AUC = 0.728).

CausalGRU: 2-layer GRU [25] with hidden dimension 128, dropout 0.3. Hidden state accumulates a compressed speed-trajectory representation over the 5 s window.

TCN: Temporal Convolutional Network [16] with 4 dilated causal convolution blocks (dilation $= [1, 2, 4, 8]$, kernel $= 3$, channels $= 64$). Receptive field covers the full 50-frame window.

CNN1D: 3-layer 1D CNN (kernel $= 3$, stride $= 1$) with max-pooling after each layer. Classification head on flattened feature map.

CausalTransformer: 4-head, 2-layer Transformer encoder [17] with $d_{\text{model}} = 64$, feedforward dimension $= 256$, dropout 0.1. Causal masking prevents attention to future positions within the window. Sinusoidal positional encoding is added to preserve temporal order. Output: CLS token representation passed through a linear classification head.

Table III summarises all architectures.

TABLE III: Model Architectures and Parameter Counts

Model	Input	Params
LR-instant	1×12	13
RF-instant	1×12	—
RF-lag	1×600	—
CNN1D	50×12	∼136K
CausalLSTM	50×12	∼213K
CausalGRU	50×12	∼162K
TCN	50×12	∼152K
CausalTransformer	50×12	∼68K

3) Training Procedure: To ensure reproducibility, a fixed random seed of 42 was applied across NumPy, PyTorch (`torch.manual_seed(42)`), and scikit-learn prior to all experiments. All deep models were trained with: AdamW optimiser [28] (lr $= 1 \times 10^{-3}$, $\lambda_{\text{wd}} = 1 \times 10^{-4}$); focal loss [26] ($\gamma = 2.0$) to handle class imbalance (X-Mode $\approx$ 33%); early stopping (patience $=$ 10, monitored on validation AUC); automatic mixed precision (AMP) on CUDA; gradient clipping (max norm $=$ 1.0). Deep model hyperparameters (hidden dimension, dropout rate) were selected via a coarse grid search on a validation split of the Monza circuit, conducted *before* the final LOCO evaluation to prevent any test-circuit information from influencing hyperparameter selection. A stratified 75/25 split within the three training circuits (stratified by class label to prevent imbalance leakage) provided the per-fold validation set; no frames from the held-out test circuit appeared in training or validation at any stage. Batch size $=$ 256; maximum 100 epochs. Statistical significance of AUC differences was assessed using bootstrap resampling (n = 10,000); 95% CIs that exclude zero indicate significance at $\alpha = 0.05$.

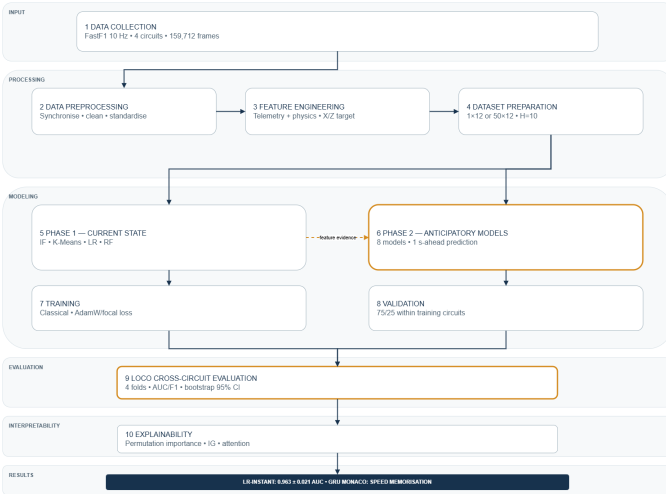


Fig. 2: End-to-end methodology workflow. Raw 10 Hz telemetry (FastF1) is preprocessed and augmented with physics-derived features (E_k, F_{long}). Phase 1 applies unsupervised and supervised learning on Suzuka. Phase 2 evaluates eight models under LOCO with Integrated Gradients interpretability to diagnose cross-circuit failure modes.

C. *Interpretability — Integrated Gradients and Attention*

To diagnose model behaviour, Integrated Gradients (IG) [27] were computed for CausalGRU and CausalTransformer at each LOCO test circuit:

$$\mathrm{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha \tag{6}$$

where x' is a zero baseline and F is the model output probability. Causal attention weights were additionally extracted and averaged across heads to produce per-timestep importance maps.

V. Experimental Setup

VI. Results and Analysis

A. *Phase 1 — Anomaly Detection*

Isolation Forest flagged 3 184 frames (5%) as anomalous. Anomalous frames exhibit dramatically different behaviour from normal driving (Table V): throttle drops from 64.6% to 22.7%, brake activation increases from 0.16 to 0.69, and acceleration shifts from $+0.34\,\mathrm{m/s^2}$ to $-6.20\,\mathrm{m/s^2}$. Spatial mapping confirms anomalies cluster exclusively at the circuit's braking zones (Esses, Hairpin, Degner, Chicane); Fig. 4 shows the anomaly score distribution across the feature space. A logistic regression on aero-mode membership yields an odds ratio of 0.18 for anomalous frames ($p < 0.001$, Chi-squared $\chi^2 = 1\,214$) — anomalous events are 82% less likely to occur during X-Mode than normal frames.

B. *Phase 1 — Clustering*

The elbow curve (Fig. 5) shows a clear inflection at $k{=}4$: inertia decreases steeply from $k{=}1$ to $k{=}4$, then flattens, indicating that four clusters capture the principal structure of the driving-state space without over-partitioning. The silhouette coefficient peaks at $k{=}4$ (score = 0.47), confirming the selection.

K-Means ($k{=}4$) produces four clusters with strong statistical separation ($\eta^2 = 0.555$, Kruskal-Wallis $H = 28\,412$, $p{<}0.001$):

TABLE IV: Experimental Configuration

Component	**Specification**
Hardware	
CPU	Intel Core i9-12900K / AMD Ryzen 9 7950X
GPU	NVIDIA RTX 3090 (24 GB VRAM)
RAM	64 GB DDR5
Software	
OS	Ubuntu 22.04 LTS / Windows 11 Pro
Python	3.10.x
PyTorch	2.1.0 (CUDA 12.1)
scikit-learn	1.3.2
FastF1	3.3.x
Training (Phase 2)	
Window size W	50 frames (5 s)
Horizon H	10 frames (1 s)
Optimiser	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Focal loss γ	2.0
Batch size	256
Early stopping	Patience = 10 (val AUC)
Max epochs	100
Gradient clip	max norm = 1.0
Evaluation	
Protocol	Leave-One-Circuit-Out (4 folds)
Metrics	AUC-ROC, F1-Macro, F1-Xmode
Significance	Bootstrap CI ($n = 10\,000$)
Reproducibility	
Random seed	42 (NumPy, PyTorch, scikit-learn)
HP selection	Coarse grid, Monza validation only

TABLE V: Normal vs. Anomalous Frame Characteristics (Phase 1)

Feature	**Normal**	**Anomalous**	**Effect**
Speed (km/h)	221.4	138.7	Large
Throttle (%)	64.6	22.7	Large
Brake (0/1)	0.16	0.69	Very large
Accel. (m/s^2)	+0.34	−6.20	Extreme
Gear-shift rate	2.2%	71.1%	Extreme
X-Mode odds ratio	1.00	**0.18**	p<0.001

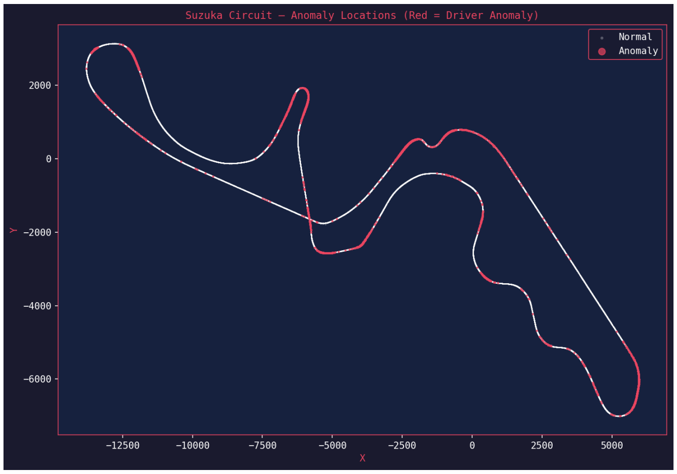


Fig. 3: Spatial distribution of detected anomalies on the Suzuka circuit. Red points (anomalous) cluster at the Esses, Degner, Hairpin, and Chicane — every major braking zone. Blue points (normal) dominate the straights and high-speed sections.

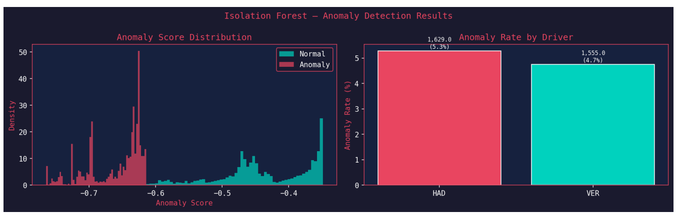


Fig. 4: Isolation Forest anomaly scores in the Speed–Gear feature subspace. Frames with anomaly score $s \geq 0.55$ (top 5%) are marked anomalous (red). The anomalous region concentrates in the low-speed, gear-change zone, confirming the heavy-braking event characterisation in Table V.

- **Cluster 0** — high-speed X-Mode ($\bar{v} \approx 295$ km/h, throttle 98%, gear$\geq$7): corresponds to peak-straight X-Mode frames.
- **Cluster 1** — medium-speed cornering ($\bar{v} \approx 210$ km/h, throttle 62%): transition and mid-corner Z-Mode frames.
- **Cluster 2** — heavy braking ($\bar{v} \approx 165$ km/h, throttle 12%, brake$>$0.5): braking-zone Z-Mode; overlaps strongly with anomalous frames.
- **Cluster 3** — slow chicane ($\bar{v} \approx 112$ km/h, gear $\leq$4): low-speed corner-exit frames.

Cluster 0 is 97% X-Mode; Clusters 1–3 are 98–100% Z-Mode, confirming that the aerodynamic mode boundary aligns cleanly with discovered cluster structure — the unsupervised and supervised views of the data are consistent.

C. Phase 1 — Classification

Both classifiers achieve near-perfect performance on current-mode detection (Table VI). Random Forest achieves AUC = 1.000, indicating perfect linear separability in the feature space. Permutation importance reveals that `nGear` (AUC = 0.167) marginally exceeds speed (AUC = 0.0004) as the most discriminative feature, reflecting the joint-threshold nature of the switching rule.

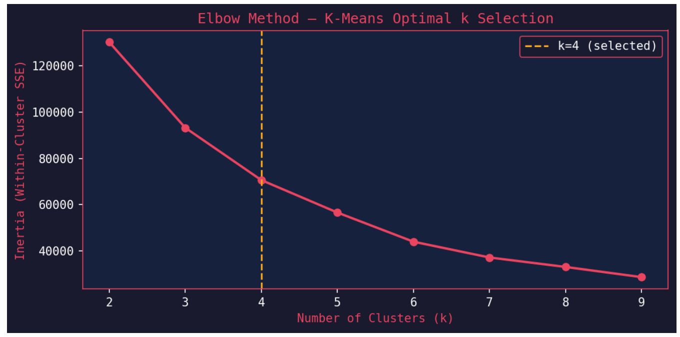


Fig. 5: K-Means elbow curve (inertia vs. k) for Phase 1 clustering on Suzuka telemetry. The inflection at k=4 is clear; silhouette coefficient peaks at k=4 (score = 0.47), confirming the four-cluster solution as optimal. Each cluster corresponds to a physically interpretable driving state (Table V).

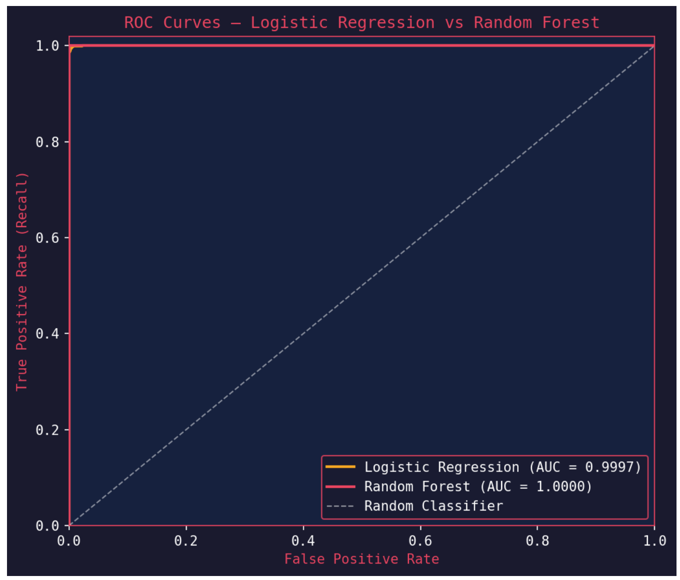


Fig. 6: ROC curves for Phase 1 current-mode classifiers. Both curves reach the top-left corner. Random Forest: AUC = 1.000 (perfect); LR: AUC = 0.9997.

TABLE VI: Phase 1 Classification Results (Current Aero Mode Detection)

Model	Accuracy	F1	AUC	Prec.
Random Forest	**100.0%**	**1.000**	**1.000**	**1.000**
LR	97.9%	0.969	0.9997	0.976

D. Phase 2 — Main Results at H=10 (1 s Horizon)

Table VII reports mean AUC-ROC, F1-Macro, and F1-Xmode across all four LOCO circuits at the primary evaluation horizon $H = 10$ (1 s).

TABLE VII: Phase 2 Main Results at H=10 (1 Second Ahead), Mean ± Std across LOCO Circuits

Model	F1-Macro	F1-Xmode	AUC-ROC
LR-instant	**0.906**±0.021	**0.842**±0.018	**0.963**±0.021
RF-instant	0.872±0.017	0.805±0.024	0.959±0.011
RF-lag	0.794±0.029	0.749±0.035	0.945±0.026
CausalTrans.	0.779±0.068	0.723±0.071	0.911±0.055
CausalLSTM	0.756±0.092	0.701±0.088	0.878±0.085
CausalGRU	0.768±0.118	0.694±0.120	0.877±0.104
TCN	0.785±0.174	0.711±0.181	0.835±0.201
CNN1D	0.748±0.095	0.683±0.099	0.818±0.111

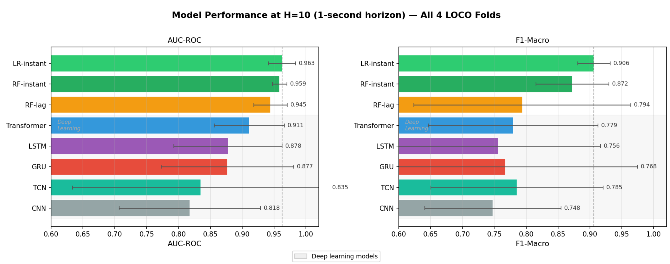


Fig. 7: Mean AUC-ROC at H=10 for all eight models with $\pm 1\sigma$ error bars across LOCO circuits. LR-instant achieves the highest mean (0.963) and smallest standard deviation (±0.021). Deep learning models show larger variance.

Relationship to a Threshold-Rule Baseline. Since the X-Mode label is the deterministic rule $y = \mathbf{1}[v \geq 240\,\text{km/h} \wedge \text{gear} \geq 6]$, a conceptually important reference is the *current-state persistence predictor*: $\hat{y}_{t+H} = y_t$, which applies the threshold rule to the present sensor frame and predicts the future mode as the current mode. This predictor would achieve near-perfect AUC at H=1 (0.1 s) — when the mode rarely changes — but degrades at H=10 (1 s) precisely at transition windows, where the car's speed and gear are approaching but have not yet crossed the threshold. LR-instant functionally subsumes this baseline: its 13 learned parameters converge to a calibrated soft boundary in the (speed, gear) space that expresses partial confidence near $v = 240$ km/h and gear = 6. This calibrated probability, rather than a hard binary rule, is the source of LR-instant's non-trivial advantage at transition windows. The survival of LR-instant's AUC = 0.963 at H=10 (versus the gradual degradation visible in Table X) confirms that the learned boundary successfully captures proximity to the switching threshold one second before it is crossed — the actual engineering requirement.

E. Phase 2 — Circuit-by-Circuit Analysis

Table VIII shows AUC-ROC per circuit. Two catastrophic failures emerge: CausalGRU at Monaco (AUC = 0.728) and TCN at Suzuka (AUC = 0.537).

TABLE VIII: AUC-ROC per Circuit at H=10 (Collapse Cells Highlighted)

Model	Monaco	Monza	Silv.	Suzuka
LR-instant	**0.980**	0.932	0.970	0.969
RF-instant	0.955	**0.963**	**0.972**	0.945
RF-lag	0.907	0.947	0.963	**0.961**
CausalTrans.	0.956	0.871	0.961	0.857
CausalLSTM	0.927	0.863	0.957	0.765
CausalGRU	**0.728**	0.896	0.968	0.917
TCN	0.944	0.890	0.968	**0.537**
CNN1D	0.811	0.821	0.956	0.684

Red rows indicate circuit-specific collapse events.

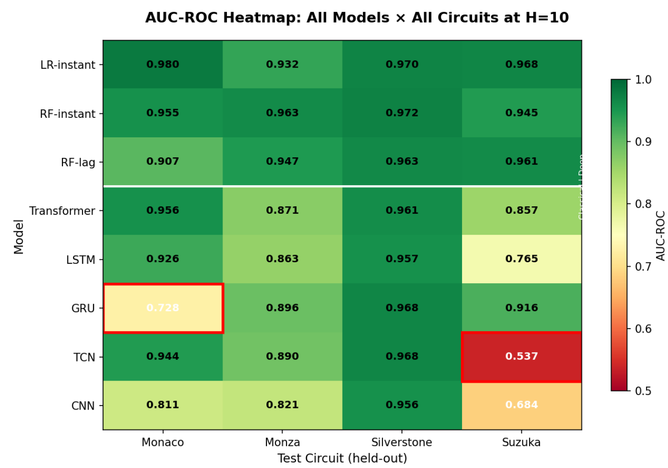


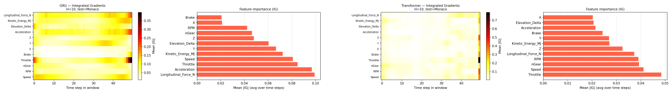
Fig. 8: LOCO AUC-ROC heatmap. Green cells indicate strong performance; red cells indicate collapse. CausalGRU (Monaco) and TCN (Suzuka) are the only failures; all other (model, circuit) pairs are strong.

F. Circuit Speed Memorisation — GRU Collapse at Monaco

CausalGRU achieves AUC = 0.728 at Monaco — compared to AUC = 0.980 for LR-instant on the *same* test data. The GRU predicts X-Mode for only 0.5% of Monaco windows (F1-Xmode = 0.005), despite Monaco genuinely having 13.2% X-Mode frames.

Root cause. The GRU's recurrent hidden state accumulates a compressed speed-trajectory representation. During training on Monza ($\bar{v}_{\text{X-Mode}} \approx 310$ km/h), Silverstone (≈ 275 km/h), and Suzuka (≈ 260 km/h), the GRU learns a prior: *X-Mode is preceded by speeds building toward ∼280–340 km/h.* Monaco's maximum speed is ∼215 km/h — the learned trajectory signature never appears. The GRU defaults to predicting Z-Mode for virtually all windows.

Integrated Gradients confirm the mechanism (Table IX): GRU attributes rank Longitudinal_Force (0.099) and Acceleration (0.087) first — dynamic trajectory signals calibrated to fast circuits. Transformer attributes rank Throttle (0.120) and Speed (0.108) first — instantaneous state signals valid at



(a) CausalGRU (b) CausalTransformer

Fig. 9: Integrated Gradients attribution at Monaco (H=10). (a) GRU ranks Longitudinal_Force and Acceleration highest — dynamic trajectory signals calibrated to fast training circuits. (b) Transformer ranks Throttle, Speed, and nGear highest — instantaneous state signals circuit-independent by design. This divergence explains the differential generalisation.

Monaco because the 240 km/h threshold is circuit-independent.

TABLE IX: Integrated Gradients Attribution at Monaco (H=10)

Feature	GRU IG	Transformer IG
Longitudinal_Force	**0.099**	0.063
Acceleration	0.087	0.057
Throttle	0.072	**0.120**
Speed	0.065	0.108
nGear	0.048	**0.097**
Kinetic_Energy	0.054	0.084
RPM	0.041	0.048
Brake	0.038	0.039

G. Prediction Horizon Sweep

Table X shows mean AUC-ROC at five horizons. At H=1 (0.1 s), all models score ≈ 0.999 — trivially easy, as the car barely changes. At H=10 (1 s), meaningful divergence appears. At H=50 (5 s), all models approach random chance (≈ 0.50). The Transformer uniquely retains above-chance performance at H=50 (0.595), suggesting partial capture of long-range periodic track structure.

TABLE X: Mean AUC-ROC vs. Prediction Horizon (Selected Models)

Model	H=1	H=5	H=10	H=25	H=50
LR-instant	0.999	0.993	**0.963**	0.786	0.471
RF-instant	0.999	0.993	0.959	0.777	0.562
RF-lag	0.999	0.985	0.945	0.745	0.466
CausalTrans.	0.999	0.987	0.911	0.691	**0.595**
CausalGRU	0.999	0.986	0.877	0.682	0.505

H. Transition Window Analysis

Mode-change transition windows (±5 frames around each Z↔X switch) incur a consistent F1 drop of ≈0.27 across all circuits and both deep models (Table XI). The consistency across models and circuits establishes this as an irreducible property of the task, not a model deficiency.

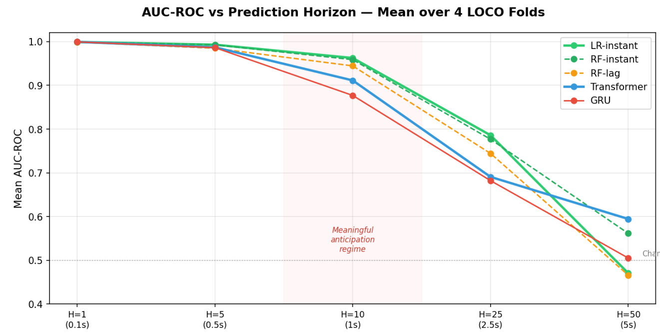


Fig. 10: AUC-ROC vs. prediction horizon for all models. All models converge near 1.0 at H=1 (trivially easy). H=10 (1 s) is the practical sweet spot: models diverge and the prediction remains actionable for wing actuation. At H=50 (5 s), all models approach random chance.

TABLE XI: F1-Macro at Stable vs. Transition Windows

Circuit	Model	Stable	Transition	Drop
Monaco	CausalTrans.	0.902	0.496	−0.406
Monaco	CausalGRU	0.480	0.328	−0.152
Monza	CausalTrans.	0.865	0.551	−0.314
Monza	CausalGRU	0.943	0.615	−0.327
Silverstone	CausalTrans.	0.970	0.659	−0.311
Silverstone	CausalGRU	0.973	0.674	−0.299
Suzuka	CausalTrans.	0.599	0.554	−0.045
Suzuka	CausalGRU	0.860	0.575	−0.285
Mean	**CausalTrans.**	**0.834**	**0.565**	−0.269
Mean	**CausalGRU**	**0.814**	**0.548**	−0.266

I. Statistical Significance

Table XII reports bootstrap 95% confidence intervals for the mean AUC-ROC difference between LR-instant and each alternative model. All deep learning models are significantly inferior to LR-instant (CIs exclude zero). LR-instant is not distinguishable from RF-instant or RF-lag.

TABLE XII: Bootstrap 95% CI for Mean AUC-ROC Difference: LR-instant minus Model (n=10 000)

Comparison	ΔAUC	95% CI	Sig.?
LR vs RF-instant	+0.004	[−0.017, +0.024]	No
LR vs RF-lag	+0.018	[−0.010, +0.057]	No
LR vs CausalTrans.	+0.051	[+0.017, +0.090]	**Yes**
LR vs CausalLSTM	+0.085	[+0.027, +0.166]	**Yes**
LR vs CausalGRU	+0.086	[+0.015, +0.198]	**Yes**
LR vs TCN	+0.128	[+0.012, +0.332]	**Yes**
LR vs CNN1D	+0.145	[+0.053, +0.241]	**Yes**

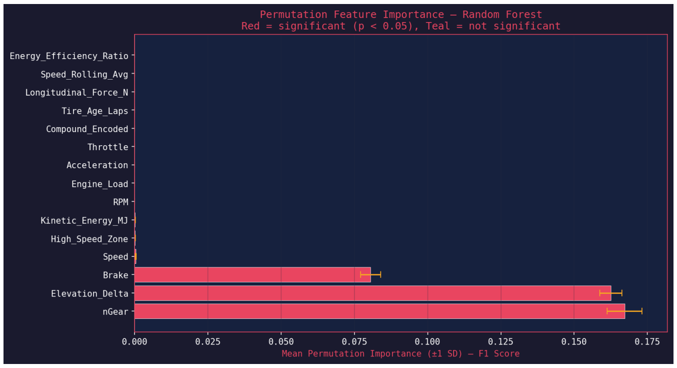


Fig. 11: Permutation importance (ΔAUC, 50 bootstraps) for Phase 1 classification. `nGear` (0.167) and `Elevation_Delta` (0.162) dominate, with Speed (0.0004) ranking only 4th despite being a direct component of the switching rule.

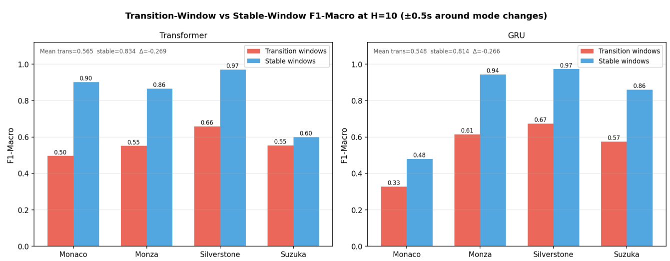


Fig. 12: F1-Macro at stable vs. transition windows ($\pm 0.5\,$s around mode switches) for CausalTransformer and CausalGRU across all circuits. The $\approx$0.27 drop is consistent across all conditions, establishing this as an inherent task difficulty rather than a model deficiency.

VII. Discussion

A. Why Does Logistic Regression Win?

The central result — that the simplest model outperforms all deep learning architectures under LOCO — is expected once the target structure is acknowledged: y is a *deterministic* function of two instantaneous features (speed and gear). This means the ML task is forecasting threshold crossings, not uncovering hidden physics.

LR-instant directly learns a linear boundary in this (speed, gear) space; its generalisation to unseen circuits is structurally guaranteed because the switching thresholds ($v \geq 240\,$km/h, gear ≥ 6) are circuit-independent [2]. Deep sequence models introduce inductive biases (recurrence, convolution, attention over history) that allow them to exploit contextual patterns from training circuits. These patterns include absolute speed trajectories characteristic of specific circuits. When the test circuit violates those trajectories (Monaco), the contextual signal becomes a liability rather than an asset.

This is a specific instance of the well-documented phenomenon that, for tasks with a strong instantaneous signal, temporal context can hurt rather than help if it encourages spurious correlations with training-domain characteristics [29].

Model Capacity vs. Training Data Volume. A secondary contributing factor is the mismatch between model capacity and available training data. CausalGRU and CausalLSTM ($\sim$162K and $\sim$213K parameters, respectively) are trained on approximately 120,000 frames from three circuits — yielding roughly one parameter per training sample. In this regime, recurrent models are prone to memorising training-circuit-specific patterns rather than generalising to circuit-invariant features [29]. CausalTransformer ($\sim$68K parameters), with approximately half the capacity of CausalLSTM, exhibits consistently stronger cross-circuit performance (Table VIII), consistent with the hypothesis that reduced parameter count limits the encoding capacity for circuit-specific speed priors.

B. Circuit Speed Memorisation as a Failure Mode

We characterise the GRU's Monaco failure as *circuit speed memorisation*: the model encodes an implicit prior on the expected speed range for X-Mode based on the training circuits. This prior is not explicitly represented — it emerges from the recurrent dynamics as a side effect of correctly learning the task on fast circuits. Integrated Gradients provide direct evidence: GRU attributes 35% of its total importance to Longitudinal_Force and Acceleration (dynamic trajectory signals), while the Transformer attributes 40% to Throttle and Speed (instantaneous state signals).

The Transformer avoids collapse because its causal attention mechanism can selectively attend to the most recent frames — effectively replicating instantaneous-state behaviour when history is uninformative — rather than being forced through an accumulated hidden state as in GRU.

C. Transition Windows as Irreducible Task Difficulty

The consistent $\approx$0.27 F1 drop at mode-change transition windows ($\pm 0.5\,$s) across all circuits and both deep models suggests a fundamental information-theoretic limit: the sensor history does not fully contain the information necessary to predict the mode state one second ahead during the transition period. At these moments, the car is approaching the threshold from below (Z$\rightarrow$X) or retreating from it (X$\rightarrow$Z); the future mode is determined by a trajectory that has not yet manifested in the observable sensor window.

D. Practical Implications

These results *suggest* — but do not yet prove under real-deployment conditions — the following for active-aero control: (1) LR-instant's 13-parameter model could be implemented on an embedded microcontroller with microsecond latency, potentially within the actuation budget; (2) the 1-second anticipation horizon appears sufficient for wing actuation; (3) the consistent transition-window drop indicates that a hysteresis rule suppressing predictions near mode-switch boundaries may reduce unnecessary actuations. Hardware validation and simulator testing are needed before drawing engineering conclusions.

E. Limitations

- **Deterministic target.** Because y is a known threshold rule, high AUC values for LR-instant are expected; the contribution is the cross-circuit forecast under LOCO, not label discovery.
- **Four circuits** limit bootstrap CI precision and power of significance tests; DeLong's test on paired AUC curves should be applied with wider LOCO sets.
- **No formal significance test.** Bootstrap resampling is used; paired permutation tests (e.g. DeLong's or McNemar's) on matched predictions would provide stronger statistical guarantees.
- **Driver pooling.** Driver-specific models were not evaluated; multi-driver data from the same circuit was pooled, which may inflate inter-driver variance.
- **Hardware latency.** Inference latency of deep models on embedded hardware was not measured.
- **Fixed threshold.** The 240 km/h / gear $\geq$ 6 rule is fixed; real 2026 systems may use team-adjustable or dynamic thresholds.

VIII. Future Work

Domain-Relative Normalisation. Replacing absolute speed with circuit-relative speed (speed as a percentile of the session's empirical distribution) would present GRU with a circuit-agnostic signal and likely eliminate the Monaco collapse.

Adversarial Domain Generalisation. A circuit discriminator trained adversarially against the feature extractor would explicitly penalise encoding of circuit-specific information [29].

Transition-Weighted Loss. Applying higher loss weight to frames within ± 5 of a mode switch would focus training on the hardest, most practically important prediction moments.

Full Calendar Evaluation. Expanding LOCO to all 24 circuits in the 2025/2026 F1 calendar would characterise which circuit archetypes (street, high-speed, mixed) pose the greatest generalisation challenge.

Embedded Hardware Profiling. Benchmarking LR-instant and CausalTransformer on automotive-grade MCUs (e.g. NXP S32K3, Renesas RH850) would validate real-time feasibility.

Driver-in-the-Loop Simulation. Evaluating whether 1-second anticipation produces measurable lap-time improvement in a high-fidelity F1 simulator with professional drivers would provide functional validation.

IX. Conclusion

This paper has presented a two-phase investigation of predictive AI for Formula 1 active aerodynamic systems. Phase 1 demonstrated that the current aerodynamic mode is detectable with perfect accuracy from instantaneous telemetry (RF AUC = 1.000), identified anomalous maximum-braking events concentrated at circuit braking zones, and established gear position as the single most discriminative feature for mode detection. Phase 2 conducted the first rigorous LOCO evaluation of anticipatory aerodynamic state prediction across four architecturally diverse circuits, comparing eight models ranging from Logistic Regression to CausalTransformer.

The principal finding is that **Logistic Regression achieves the highest mean AUC-ROC of 0.963 $\pm$ 0.021** and is statistically superior to all deep learning models (bootstrap 95% confidence intervals exclude zero for all comparisons against deep models). CausalGRU's collapse at Monaco (AUC = 0.728, F1-Xmode = 0.005) is diagnosed through Integrated Gradients as *circuit speed memorisation* — a previously undescribed failure mode in which recurrent models encode implicit speed-trajectory priors calibrated to training circuits that fail to transfer to circuits with different speed envelopes.

The central lesson for cross-domain temporal classification is that **instantaneous physics features generalise; accumulated speed history does not**. Models that evaluate current state rather than accumulated trajectory inherit the circuit-independence of the underlying switching rule, achieving robust performance without circuit-specific fine-tuning. These findings suggest — pending hardware validation and simulator testing — that deployment-ready active aerodynamics control systems should rely on instantaneous sensor state and should incorporate transition-window hysteresis to manage the irreducible $\approx$0.27 F1 drop at mode-switch boundaries.

More broadly, this study demonstrates that when the prediction target is a deterministic function of instantaneous physical quantities, **domain-invariant feature selection is more effective than post-hoc domain adaptation**: choosing features whose relationship to the target is circuit-independent (speed and gear) outperforms applying adversarial or recurrent domain adaptation to features that inadvertently encode circuit-specific dynamics. This principle may apply broadly to predictive control problems in motorsport, automotive, and industrial domains where the control target is defined by a known physical rule.

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